

# Chapter 7: Logarithmic and Exponential Functions

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## 7.1 Inverse Function

### Derivative of the Inverse Function

Let  $f$  be differentiable and have an inverse on an interval  $I$ . If  $x_0$  is a point of  $I$  at which  $f'(x_0) \neq 0$ , then  $f^{-1}$  is differentiable at  $y_0 = f(x_0)$  and

$$(f^{-1})'(y_0) = \frac{1}{f'(x_0)}, \text{ where } y_0 = f(x_0) \quad (1)$$

## 7.2 The Natural Logarithmic and Exponential Functions

### The Natural Logarithm

The **natural logarithm** of a number  $x > 0$ , denoted  $\ln x$ , is defined

$$\ln x = \int_1^x \frac{dt}{t} \quad (2)$$

### Properties of the Natural Logarithm

1. The domain and range of  $\ln x$  are  $(0, \infty)$  and  $(-\infty, \infty)$ , respectively.
2.  $\ln(xy) = \ln x + \ln y, \forall x, y \in \mathbf{R}^+$
3.  $\ln(x/y) = \ln x - \ln y, \forall x, y \in \mathbf{R}^+$
4.  $\ln x^p = p \ln x, \forall x \in \mathbf{Q}^+$
5.  $\frac{d}{dx}(\ln |x|) = \frac{1}{x}, \forall x \in \mathbf{R} \wedge x \neq 0$
6.  $\frac{d}{dx}(\ln |u(x)|) = \frac{u'(x)}{u(x)}$
7.  $\int \frac{dx}{x} = \ln |x| + C$

### The Number $e$

The number  $e$  is the real number that satisfies

$$\ln e = \int_1^e \frac{dt}{t} = 1 \quad (3)$$

### The Exponential Function

$\forall x, y \in \mathbf{R}$

$$y = e^x \Leftrightarrow x = \ln y \quad (4)$$

### Properties of $e$

- $e^{x+y} = e^x e^y$
- $e^{x-y} = e^x / e^y$
- $(e^x)^y = e^{xy}, \forall y \in \mathbf{Q}$
- $\ln(e^x) = x, \forall x \in \mathbf{R}$
- $e^{\ln x} = x, \forall x \in \mathbf{R}^+$

### Exponential Functions with General Bases

Let  $b \in \mathbf{R}^+ \wedge b \neq 1. \forall x \in \mathbf{R}^+$ ,

$$b^x = e^{x \ln b} \quad (5)$$

### Derivative and Integral of the Exponential Function

$\forall x \in \mathbf{R}$ ,

$$\frac{d}{dx}(e^{u(x)}) = e^{u(x)} u'(x) \quad (6)$$

$$\int e^x dx = e^x + C \quad (7)$$

## 7.3 Logarithmic and Exponential Functions with Other Bases

### Logarithmic Function Base $b$

For any base  $b > 0$ , with  $b \neq 1$ , the **logarithmic function base  $b$** , denoted  $\log_b x$ , is the inverse of the exponential function  $b^x$ .

### Inverse Relations for Exponential and Logarithmic Functions

For any base  $b > 0$ , with  $b \neq 1$ , the following inverse relation holds.

- $b^{\log_b x} = x, \forall x \in \mathbf{R}^+$
- $\log_b b^x = x, \forall x$

### Derivative of $b^x$

If  $b > 0 \wedge b \neq 1, \forall x$ ,

$$\frac{d}{dx}(b^x) = b^x \ln b \quad (8)$$

### Indefinite Integral of $b^x$

For  $b > 0 \wedge b \neq 1$ ,

$$\int b^x dx = \frac{1}{\ln b} b^x + C \quad (9)$$

### General Power Rule

$\forall p, x \in \mathbf{R}^+$ ,

$$\frac{d}{dx}(x^p) = px^{p-1} \quad (10)$$

Furthermore, if  $u$  is a positive differentiable function on its domain, then

$$\frac{d}{dx}(u(x)^p) = p(u(x))^{p-1} \times u'(x) \quad (11)$$

### Derivative of $\log_b x$

If  $b > 1$ ,

$$\frac{d}{dx}(\log_b x) = \frac{1}{x \ln b} \quad \forall x \in \mathbf{R}^+ \wedge x \neq 0 \quad (12)$$

$$\frac{d}{dx}(\log_b |x|) = \frac{1}{x \ln b} \quad \forall x \in \mathbf{R}^+ \wedge x \neq 0 \quad (13)$$

## 7.5 Inverse Trigonometric Functions

### Derivative of Inverse Trigonometric Functions

$$\begin{aligned}\frac{d}{dx}(\sin^{-1} x) &= \frac{1}{\sqrt{1-x^2}} & \frac{d}{dx}(\cos^{-1} x) &= -\frac{1}{\sqrt{1-x^2}}, \{x \in \mathbf{R} \mid -1 < x < 1\} \\ \frac{d}{dx}(\tan^{-1} x) &= \frac{1}{1+x^2} & \frac{d}{dx}(\cot^{-1} x) &= -\frac{1}{1+x^2}, \{x \in \mathbf{R} \mid -\infty < x < \infty\} \\ \frac{d}{dx}(\sec^{-1} x) &= \frac{1}{|x|\sqrt{x^2-1}} & \frac{d}{dx}(\csc^{-1} x) &= -\frac{1}{|x|\sqrt{x^2-1}}, \{x \in \mathbf{R} \mid |x| > 1\}\end{aligned}$$

### Integrals Involving Inverse Trigonometric Functions

$$\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} \frac{x}{a} + C \quad (14)$$

$$\int \frac{dx}{a^2 + x^2} = \frac{1}{a} \tan^{-1} \frac{x}{a} + C \quad (15)$$

$$\int \frac{dx}{x\sqrt{a^2 + x^2}} = \frac{1}{a} \sec^{-1} \left| \frac{x}{a} \right| + C \quad (16)$$

## 7.6 L'Hôpital's Rule and Growth Rates of Functions

**Indeterminate forms**  $1^\infty, 0^0, \infty^0$

Assume  $\lim_{x \rightarrow \infty} f(x)^{g(x)}$  has the indeterminate form  $1^\infty, 0^0$ , or  $\infty^0$ .

1. Evaluate  $L = \lim_{x \rightarrow \infty} g(x) \ln f(x)$ . This limit can be put in the form  $0/0$  or  $\infty/\infty$ , both of which are handled by l'Hôpital's rule.
2. Then  $\lim_{x \rightarrow \infty} f(x)^{g(x)} = e^L$

**Growth Rates of Functions (as  $x \rightarrow \infty$ )**

Suppose  $f$  and  $g$  are functions with  $\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} g(x) = \infty$ . Then  $f$  **grows faster than  $g$**  as  $x \rightarrow \infty$  if

$$\lim_{x \rightarrow \infty} \frac{g(x)}{f(x)} = 0 \quad \text{or, quantitatively, if} \quad \lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \infty \quad (17)$$

The functions  $f$  and  $g$  have **comparable growth rates** if

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = M$$

where  $M \in \mathbb{R}^+$ .

**Ranking Growth Rates as  $x \rightarrow \infty$**

Let  $f \ll g$  mean that  $g$  grows faster than  $f$  as  $f \rightarrow \infty$ . With positive real numbers  $p, q, r, s$  and  $b > 1$ ,

$$\ln^q x \ll x^p \ll x^p \ln^r x \ll x^{p+s} \ll b^x \ll x^x \quad (18)$$



## 7.7 Hyperbolic Functions

### Hyperbolic Functions

#### Hyperbolic Cosine

$$\cosh x = \frac{e^x + e^{-x}}{2} \quad (19)$$

#### Hyperbolic Sine

$$\sinh x = \frac{e^x - e^{-x}}{2} \quad (20)$$

#### Hyperbolic Tangent

$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}} \quad (21)$$

#### Hyperbolic Cotangent

$$\coth x = \frac{\cosh x}{\sinh x} = \frac{e^x + e^{-x}}{e^x - e^{-x}} \quad (22)$$

#### Hyperbolic Secant

$$\operatorname{sech} x = \frac{1}{\cosh x} = \frac{2}{e^x + e^{-x}} \quad (23)$$

#### Hyperbolic Cosecant

$$\operatorname{csch} x = \frac{1}{\sinh x} = \frac{2}{e^x - e^{-x}} \quad (24)$$

### Hyperbolic Identities

$$\cosh^2 x - \sinh^2 x = 1$$

$$1 - \tanh^2 x = \operatorname{sech}^2 x$$

$$\coth^2 x - 1 = \operatorname{csch}^2 x$$

$$\coth(x + y) = \cosh x \cosh y + \sinh x \sinh y$$

$$\sinh(x + y) = \sinh x \cosh y + \cosh x \sinh y$$

$$\cosh 2x = \cosh^2 x + \sinh^2 x$$

$$\cosh^2 x = \frac{\cosh 2x + 1}{2}$$

$$\cosh(-x) = \cosh x$$

$$\sinh(-x) = -\sinh x$$

$$\tanh(-x) = -\tanh x$$

$$\sinh 2x = 2 \sinh x \cosh x$$

$$\sinh^2 x = \frac{\cosh 2x - 1}{2}$$

## Derivatives and Integral Formulas

1.  $\frac{d}{dx}(\cosh x) = \sinh x \Rightarrow \int \sinh x \, dx = \cosh x + C$
2.  $\frac{d}{dx}(\sinh x) = \cosh x \Rightarrow \int \cosh x \, dx = \sinh x + C$
3.  $\frac{d}{dx}(\tanh x) = \operatorname{sech}^2 x \Rightarrow \int \operatorname{sech}^2 x \, dx = \tanh x + C$
4.  $\frac{d}{dx}(\coth x) = -\operatorname{csch}^2 x \Rightarrow \int \operatorname{csch}^2 x \, dx = -\coth x + C$
5.  $\frac{d}{dx}(\operatorname{sech} x) = -\operatorname{sech} x \tanh x \Rightarrow \int \operatorname{sech} x \tanh x \, dx = -\operatorname{sech} x + C$
6.  $\frac{d}{dx}(\operatorname{csch} x) = -\operatorname{csch} x \coth x \Rightarrow \int \operatorname{csch} x \coth x \, dx = -\operatorname{csch} x + C$

## Integrals of Hyperbolic Functions

1.  $\int \tanh x \, dx = \ln \cosh x + C$
2.  $\int \coth x \, dx = \ln |\sinh x| + C$
3.  $\int \operatorname{sech} x \, dx = \tan^{-1} |\sinh x| + C$
4.  $\int \operatorname{csch} x \, dx = \ln |\tanh(x/2)| + C$

## Inverses of the Hyperbolic Functions Expressed as Logarithms

$$\begin{aligned}\cosh^{-1} x &= \ln(x + \sqrt{x^2 - 1}) \quad (x \geq 1) & \operatorname{sech}^{-1} x &= \cosh^{-1} \frac{1}{x} \quad (0 < x \leq 1) \\ \sinh^{-1} x &= \ln(x + \sqrt{x^2 + 1}) & \operatorname{csch}^{-1} x &= \sinh^{-1} \frac{1}{x} \quad (x \neq 0) \\ \tanh^{-1} x &= \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right) \quad (|x| < 1) & \coth^{-1} x &= \tanh^{-1} \frac{1}{x} \quad (|x| > 1)\end{aligned}$$

## Derivatives of the Inverse Hyperbolic Functions

$$\begin{aligned}\frac{d}{dx}(\cosh^{-1} x) &= \frac{1}{\sqrt{x^2 - 1}} \quad (x > 1) & \frac{d}{dx}(\sinh^{-1} x) &= \frac{1}{\sqrt{x^2 + 1}} \\ \frac{d}{dx}(\tanh^{-1} x) &= \frac{1}{1 - x^2} \quad (|x| < 1) & \frac{d}{dx}(\coth^{-1} x) &= \frac{1}{1 - x^2} \quad (|x| > 1) \\ \frac{d}{dx}(\operatorname{sech}^{-1} x) &= -\frac{1}{x\sqrt{1 - x^2}} \quad (0 < x < 1) & \frac{d}{dx}(\operatorname{csch}^{-1} x) &= -\frac{1}{|x|\sqrt{1 + x^2}} \quad (x \neq 0)\end{aligned}$$

## Integral Formulas

1.  $\int \frac{dx}{\sqrt{x^2 - a^2}} = \cosh^{-1} \frac{x}{a} + C, \text{ for } x > a$
2.  $\int \frac{dx}{x^2 + a^2} = \sinh^{-1} \frac{x}{a} + C, \text{ for all } x$

3.  $\int \frac{dx}{a^2-x^2} = \frac{1}{a} \tanh^{-1} \frac{x}{a} + C$ , for  $|x| < a = \frac{1}{a} \coth^{-1} \frac{x}{a} + C$ , for  $|x| > a$
4.  $\int \frac{dx}{x\sqrt{a^2-x^2}} = -\frac{1}{a} \operatorname{sech}^{-1} \frac{x}{a} + C$ , for  $0 < x < a$
5.  $\int \frac{dx}{x\sqrt{a^2+x^2}} = -\frac{1}{a} \operatorname{csch}^{-1} \frac{|x|}{a} + C$ , for  $x \neq 0$